Fermilab | U.S. DEPARTMENT OF ENERGY | Office of Science

Computer Vision and Cavity Cutouts

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SRF Weekly Meeting

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In partnership with:

THE UNIVERSITY OF CHICAGO

Motivation

Nb-H compounds increase surface resistance, hence
- decreases Q_0 of cavity, particularly at high values of E_{acc}.
- decreases usable range of E_{acc}.

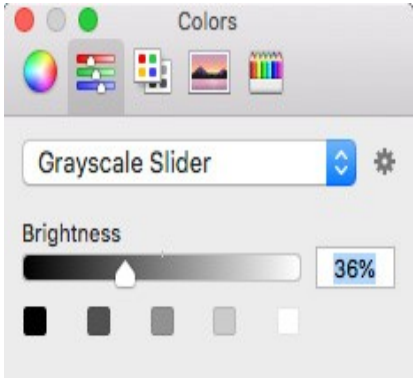
Main set of data is LCM images of cavity cutouts, which we want to understand at a deeper level than what a visual inspection would give.

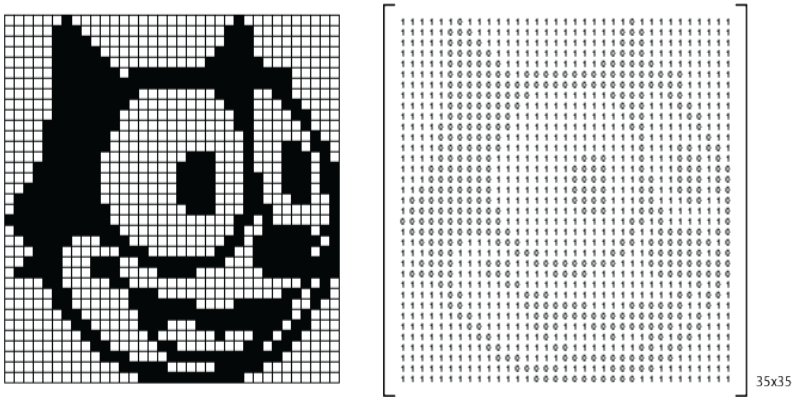
In particular, we want to understand the differences in cooldown/ warmup behavior when the sample is modified: e.g. electron irradiation, N doping, H loading, etc.

Interpreting Images

To us, images are purely visual media, but to a computer, an image is a matrix of numbers in [0,255], where 0 is black and 255 is white.

The task at hand is for computers to recover the visual data we process with our eyes.





(Klein Digital Blog)

Mathematical Preliminaries

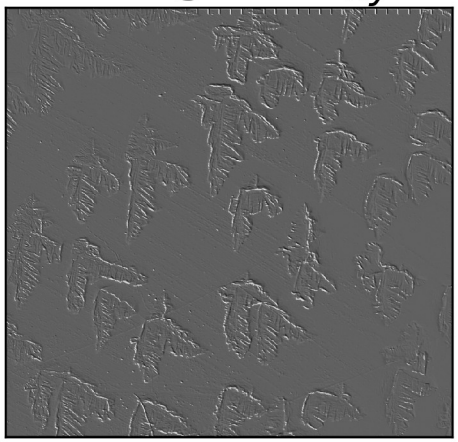
- Image: Function $f : [m] \times [n] \to [0, 255]$
- Discrete Gradient
  - $\Delta_x f(i, j; h) = \dfrac{f(i+h, j) - f(i, j)}{h}$
  - $\Delta_y f(i, j; h) = \dfrac{f(i, j+h) - f(i, j)}{h}$
  - $|\nabla f| = \sqrt{(\Delta_x f)^2 + (\Delta_y f)^2}$ $\longrightarrow$ Edges in Image / Precipitates on sample
- Spatial Frequency
  - Fourier conjugates of position: $\nu_x, \nu_y \longleftrightarrow x, y$ [1/m]
- Convolution
  - $(f \star g)(x, y) = \int_{\mathbb{R}^2} f(a, b) g(x - a, y - b) da db$
  - Acts as a “filtration” process for images: Edge detection, smoothing,...

Pictures!

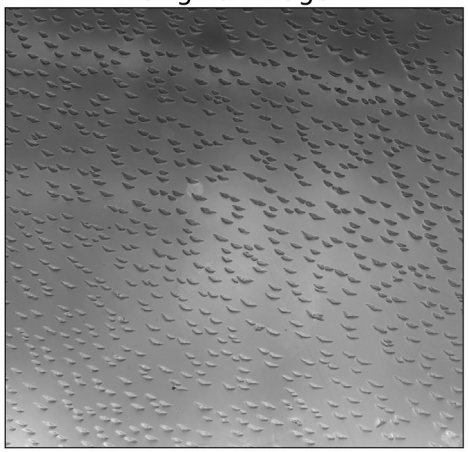
Original Image

$$\star h_{xy} = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

Image * h_{xy}

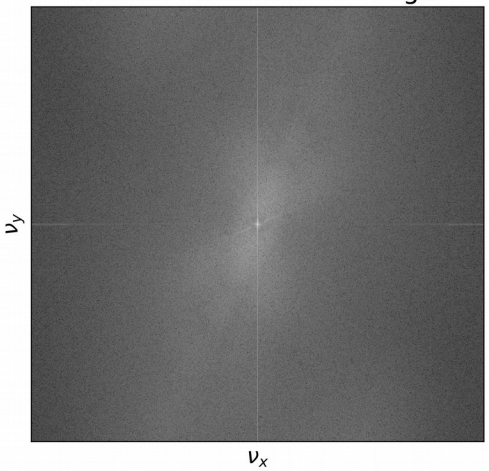


Original Image



2D FFT

Fourier Transform of Image



Attempt 1: Discrete Gradient Magnitude

Discrete Grad Gif Here.

$$|\nabla f| = \sqrt{(\Delta_x f)^2 + (\Delta_y f)^2}$$

Pros:

1. Intuitive

2. Clear visual connection between picture and analysis

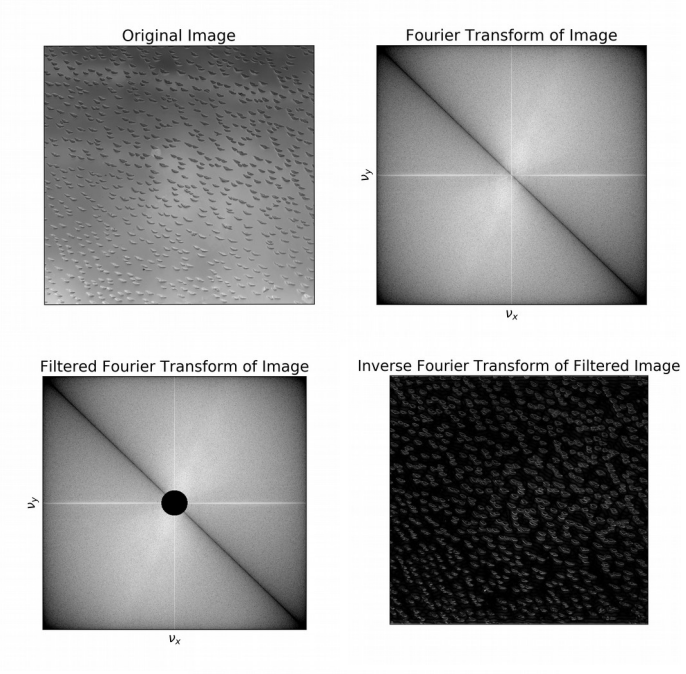
Cons:

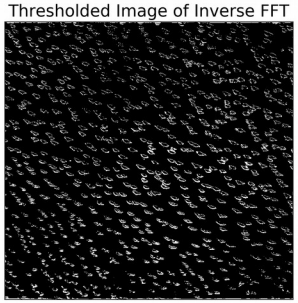
1. Hard to find a physical parameter from the edge magnitude

Attempt 2: Fourier Analysis

Goal: Determine how much of image is occupied by edges/high-frequency data

1. Fourier Transform image.
2. Filter out low-frequency data
   - This isolates the part of the image that is dominated by small features
3. Inverse Fourier Transform image.
4. Threshold Resulting Image
   - This separates the high-frequency points from the low-frequency points to count them later.





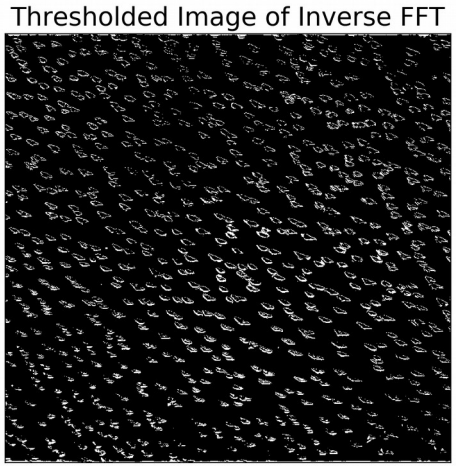
Attempt 2: Fourier Analysis

Result: Filling fraction.

$$\text{Filling Fraction} = \frac{\#\ \text{White Pixels}}{\#\ \text{Total Pixels}}$$

$$= \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n} M_{i,j}$$

M =



This gives a crude (normalized) estimate to how much of the sample is deformed due to Nucleation centers for Nb-H compounds.

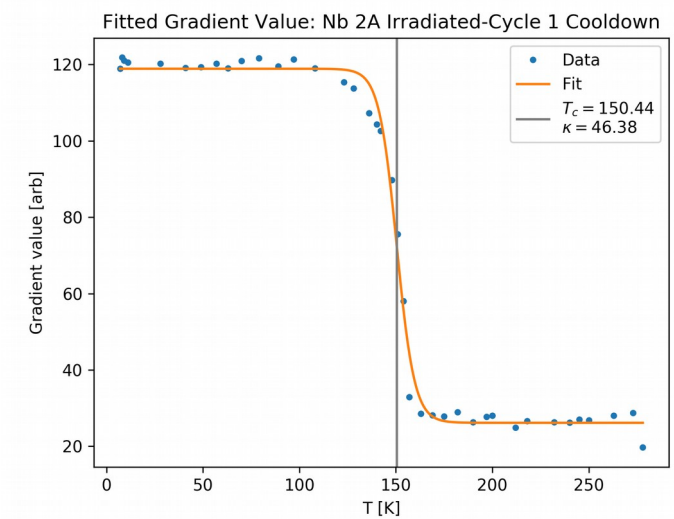
Filling Fraction vs Temperature

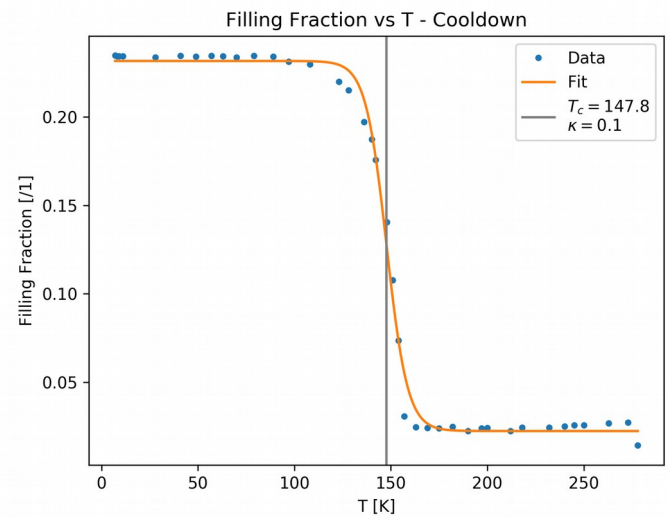
Filling Fraction Gif Here.

$$\mathrm{F.F.} = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n} M_{i,j}$$

Note: This looks very similar to the gradient vs temperature plot, so we seem to be detecting the same effect!

DC Superconductivity Analogy

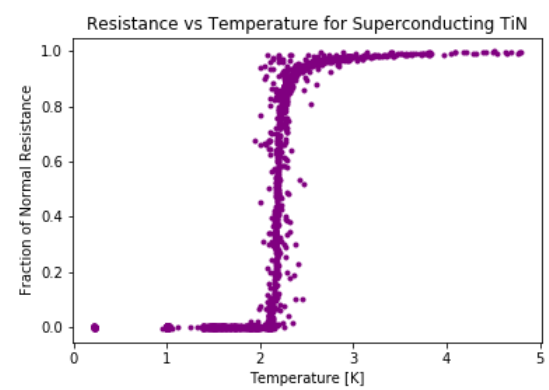




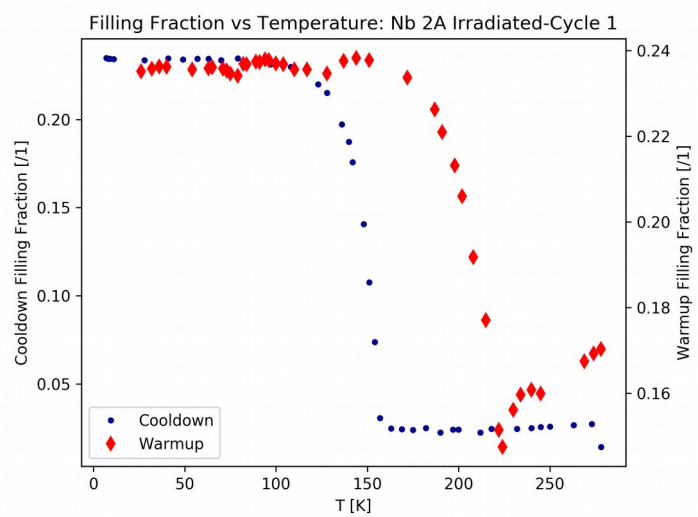
These look like resistivity vs T plots from DC superconductivity, so let's treat them as such! We fit to the function

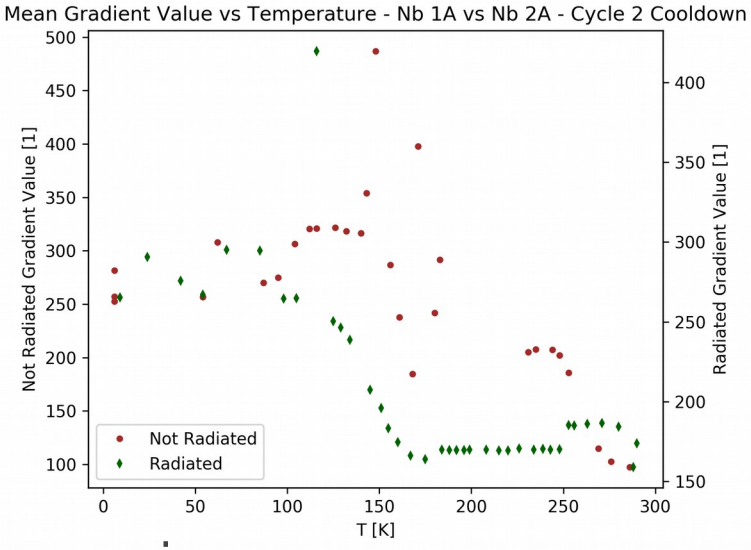
$$R = A \arctan(\kappa(T_c - T)) + B$$

, extracting the critical temperature and a parameter closely related to the transition width.



Using Computer Vision to Compare Samples





Let's put this framework to use with 2 comparisons:

1. Compare one sample's cooldown and warmup behavior
2. Compare samples in the same environment before and after irradiation

Note: This analysis is very sensitive to image acquisition details, i.e. moving the camera, sudden increases or decreases in exposure, …

To Do: Develop a standard image processing sequence on the microscope, so the data is as consistent as possible.

Questions? | Shameless Plug

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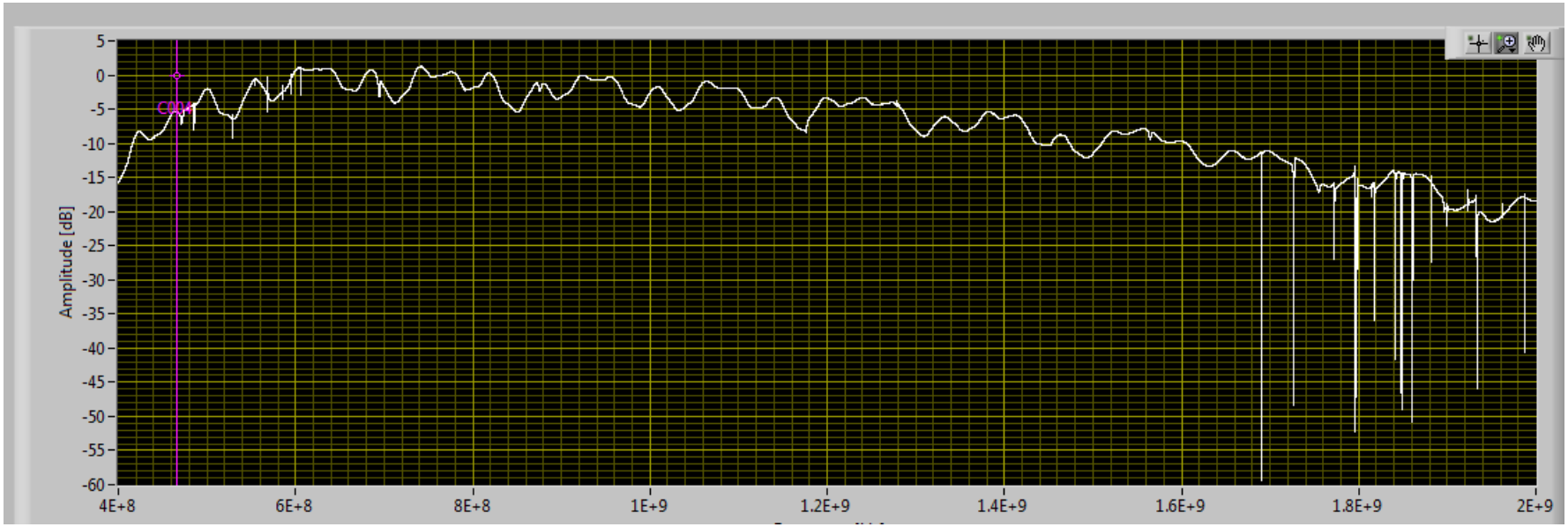
Resume and Research Experience on Website.